Shreelekha Revankar

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EDUCATION

Cornell University, Ithaca, NY **2023 - Present**
Ph.D. in Computer Science
Advised by Prof. Kavita Bala and Prof. Bharath Hariharan

University of Maryland, College Park, MD **2021 - 2022**
M.S. in Computer Science
Advised by Prof. Ming Lin as part of the combined BS/MS program

University of Maryland, College Park, MD **2017 - 2021**
B.S. in Computer Science

RESEARCH INTERESTS

Developing scalable representation learning and multi-modal reasoning techniques for unstructured data, with a specific focus on Earth observation.

SELECTED PUBLICATIONS

Revankar, S., Klassen, M., Hoover, A., Levit, C., Bala, K., Hariharan, B. "Needle in Petabyte Sized Haystack: Event Prediction and Retrieval at a Global Scale" - *In Progress*

Revankar, S., Phoo, C.P., Mall, U., Hariharan, B., Bala, K. "Scale-Aware Recognition in Satellite Images under Resource Constraints" - *ICLR 2025*

Revankar, S., Phoo, C.P., Mall, U., Bala, K., Hariharan, B. "MONITRS: Multimodal Observations of Natural Incidents Through Remote Sensing" - *NeurIPS Datasets & Benchmarks 2025* **(Spotlight, Top 2.8%)**

Kao, C.H., Zhao, W., **Revankar, S.**, et al. "Towards LLM Agents for Earth Observation" - *ARR 2026* (Submitted)

Zheng, L.Y., Wei, W., Wu, T., Clements, J., **Revankar, S.**, Harrison, A., Shen, Y., Lin, M. "Adaptive Sensitivity Analysis for Robust Augmentation against Natural Corruptions in Image Segmentation" - *ICML 2025*

WORK EXPERIENCE

Planet Labs - AI/ML Research Engineer **2025 - Present**
San Francisco, CA
- Built RAG pipeline for PlanetView (satellite imagery MLLM): semantic search, change detection, event retrieval, automatic relevance feedback
- Created Geospatial & Earth Observation skills for real-time environmental monitoring (fires, earthquakes, floods, disasters)
- Developed global construction progress monitoring using satellite embeddings to track building phases across time

Planet Labs - AI/ML Research Intern **2024**
San Francisco, CA
- Internship led to full-time offer and Planet Labs PhD Fellowship (second recipient of Planet's full PhD funding)

Shield AI - Behavior Engineering Intern **2023**
Rosslyn, VA
- Developed novel search and relocation algorithms enabling coordinated swarm behavior in aerial vehicles without inter-vehicle communication, allowing for hivemind-like operations through decentralized local rules

Bowers CIS Research - Graduate Research Mentor **2024 - 2025**
Ithaca, NY
- Mentored two undergraduates (now pursuing PhD at USC and MS at Stanford) on a vision-language foundation model for satellite imagery analysis

HONORS & INVITED TALKS

- Planet Labs PhD Fellowship - Second recipient of Planet's full PhD funding — 2025 - Present
- NeurIPS Datasets & Benchmarks 2025 Spotlight - Top 2.8% of 1,995 submissions — 2025
- Invited Talk: "Intelligent Selection of Resolution for Recognition" - AI Climate Institute — 2024
- Outstanding TA Award, nominated by Prof. Anne Bracy — 2024

SERVICE

- Invited Conference Reviewer: CVPR 2026, ICML 2026, ECCV 2026 — 2025 - Present
- Judge, Cornell Undergraduate Research Board Spring Symposium — 2024
- Graduate Mentor, Software-defined Network Interface (SONIC) Workshop — 2024

TEACHING

- Graduate TA, Cornell University: CS 1110 Introduction to Computing in Python — 2023 - 2024

SKILLS

Programming: Python, Java, C++, Kotlin, Ruby, OCaml, C
AI: MCP Servers, Agentic Workflows, LLM Agents, RAG Systems
Miscellaneous: Google Earth Engine, Geospatial Computation, BigQuery, Spanner, Firebase